

SOVITH KUMAR SINGH

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EDUCATION

Indian Institute Of Technology Delhi (2023-2025)

M. Sc. In Mathematics - CGPA - 6.00

Techno Mission International School (2018-2020)

Class 12th

Deshbandhu College, University of Delhi (2020 – 2023)

B. Sc. In Mathematics - CGPA - 8.824

De Nobili School (ICSE) (2017-2018)

Class 10th

PROJECTS

Conversational AI Chatbot with LangGraph & Streamlit

Python · LangChain · LangGraph · Google Gemini · SQLite · Streamlit

July 2025 – Sep 2025

- Engineered a stateful multi-session AI chatbot using **LangGraph StateGraph** with **SQLite checkpointing** and **Google Gemini (gemini-1.5-flash)** with tool-calling support, enabling persistent conversation history and seamless session recovery across interruptions.
- Built a **Streamlit frontend** with **multi-thread conversation management**, real-time response streaming, and a modular backend using **ToolNode**, conditional edges, and message aggregation with **LangGraph**.

Laptop Price Prediction — End-to-End ML Regression Pipeline

Python · Pandas · Numpy · Scikit-Learn · Matplotlib · Seaborn

Jan 2025 – Feb 2025

- Achieved **$R^2 = 0.896$** and **$MAE \approx 0.15$** (log-price scale) using **XGBoost** across **1,300+ records**, identifying SSD capacity, PPI, CPU class, and GPU brand as dominant pricing factors.
- Evaluated **10+ models** including Random Forest, Gradient Boosting, and SVR using an automated pipeline with **ColumnTransformer**, **One-Hot Encoding**, and log-price transformation.

TECHNICAL SKILLS

Languages & Databases: Python, SQL, PostgreSQL, MySQL, MongoDB, Snowflake

Python (Libraries & Frameworks): NumPy, Pandas, Matplotlib, Seaborn, Scikit-Learn, TensorFlow, PyTorch, LangChain, LangGraph, LangSmith, PySpark

Machine Learning (Algorithms): Regression (Simple Linear, Multiple Linear, and Polynomial), Regularization (Ridge, Lasso, and ElasticNet), Classification (Logistic and Multinomial Logistic), Decision Trees, Random Forest, Boosting (AdaBoost, Gradient, XGBoost)

Deep Learning (Algorithms): Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN)

LLM & Generative AI: Transformers, LangChain, LangGraph, LangSmith

Data Engineering & Cloud: Snowflake, PySpark, Docker, MongoDB, PostgreSQL, MySQL

Tools & Software: GitHub, Jupyter, Google Colab, MATLAB, Wolfram Mathematica, LaTeX

CERTIFICATES

- Data Analytics Projects Certification – CampusX